



C Language – 100 Questions with Solutions



LEVEL 1 – BASICS (1–20)

Concepts: Variables, Data Types, Input/Output, Operators

1. Print "Hello World"

```
#include <stdio.h>
int main() {
    printf("Hello World");
    return 0;
}
```

Hello World

2.Add two numbers

```
#include <stdio.h>
int main() {
    int a, b, sum;
    printf("Enter two numbers: ");
    scanf("%d%d", &a, &b);
    sum = a + b;
    printf("Sum = %d", sum);
    return 0;
}
```

Enter two numbers: 5 7
Sum = 12

Find area of a circle

```
#include <stdio.h>
int main() {
    float r, area;
    printf("Enter radius: ");
    scanf("%f", &r);
    area = 3.14 * r * r;
    printf("Area = %.2f", area);
    return 0;
}
```

Enter radius: 5
Area = 78.50

4 Swap Two Numbers (Using Third Variable)

```
#include <stdio.h>
int main() {
    int a, b, temp;
    printf("Enter two numbers: ");
    scanf("%d%d", &a, &b);
    temp = a;
    a = b;
    b = temp;
    printf("After swapping: a = %d, b = %d", a, b);
    return 0;
}
```

Enter two numbers: 10 20

After swapping: a = 20, b = 10

5 Even or Odd Number

```
#include <stdio.h>
int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n % 2 == 0)
        printf("Even Number");
    else
        printf("Odd Number");
    return 0;
}
```

Enter a number: 7
Odd Number

6 Largest of Two Numbers

Enter two numbers: 8 15
15 is larger

```
#include <stdio.h>
int main() {
    int a, b;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    if (a > b)
        printf("%d is larger", a);
    else
        printf("%d is larger", b);
    return 0;
}
```

7 Largest of Three Numbers

```
#include <stdio.h>
int main() {
    int a, b, c;
    printf("Enter three numbers: ");
    scanf("%d%d%d", &a, &b, &c);
    if (a >= b && a >= c)
        printf("%d is largest", a);
    else if (b >= a && b >= c)
        printf("%d is largest", b);
    else
        printf("%d is largest", c);
    return 0;
}
```

Enter three numbers: 12 45 30
45 is largest

8 Check Positive, Negative or Zero

```
#include <stdio.h>
int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n > 0)
        printf("Positive");
    else if (n < 0)
        printf("Negative");
    else
        printf("Zero");
    return 0;
}
```

Enter a number: -9
Negative

9 Sum of First N Natural Numbers (Loop)

```
#include <stdio.h>
int main() {
    int n, sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
        sum += i;
    }
    printf("Sum = %d", sum);
    return 0;
}
```

Enter n: 5
Sum = 15

10 Factorial of a Number

Enter a number: 5
Factorial = 120

```
#include <stdio.h>
int main() {
    int n, fact = 1;
    printf("Enter a number: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
        fact *= i;
    }
    printf("Factorial = %d", fact);
    return 0;
}
```

1 1 Check Leap Year

```
#include <stdio.h>
```

```
int main() {
```

```
    int year;
```

```
    printf("Enter a year: ");
```

```
    scanf("%d", &year);
```

```
    if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0))
```

```
        printf("%d is a Leap Year", year);
```

```
    else
```

```
        printf("%d is not a Leap Year", year);
```

```
    return 0;
```

```
}
```

Enter a year: 2024

2024 is a Leap Year

1 **2** Simple Interest

Enter Principal, Rate, Time: 1000 5 2
Simple Interest = 100.00

```
#include <stdio.h>
int main() {
    float p, r, t, si;
    printf("Enter Principal, Rate, Time: ");
    scanf("%f %f %f", &p, &r, &t);

    si = (p * r * t) / 100;

    printf("Simple Interest = %.2f", si);
    return 0;
}
```

1 3 Compound Interest

```
#include <stdio.h>
#include <math.h>
int main() {
    float p, r, t, ci;
    printf("Enter Principal, Rate, Time: ");
    scanf("%f %f %f", &p, &r, &t);

    ci = p * (pow((1 + r / 100), t) - 1);

    printf("Compound Interest = %.2f", ci);
    return 0;
}
```

Enter Principal, Rate, Time: 1000 10 2
Compound Interest = 210.00

1 4 ASCII Value of Character

```
#include <stdio.h>
int main() {
    char ch;
    printf("Enter a character: ");
    scanf("%c", &ch);
    printf("ASCII value of %c = %d", ch, ch);
    return 0;
}
```

Enter a character: A
ASCII value of A = 65

1 5 Convert Celsius to Fahrenheit

```
#include <stdio.h>
int main() {
    float c, f;
    printf("Enter temperature in Celsius: ");
    scanf("%f", &c);

    f = (c * 9 / 5) + 32;

    printf("Temperature in Fahrenheit = %.2f", f);
    return 0;
}
```

Enter temperature in Celsius: 37
Temperature in Fahrenheit = 98.60

1 6 Find Remainder and Quotient

```
#include <stdio.h>
int main() {
    int a, b, quotient, remainder;
    printf("Enter dividend and divisor: ");
    scanf("%d %d", &a, &b);

    quotient = a / b;
    remainder = a % b;

    printf("Quotient = %d\n", quotient);
    printf("Remainder = %d", remainder);
    return 0;
}
```

Enter dividend and divisor: 17 5
Quotient = 3
Remainder = 2

1 7 Power of a Number

```
#include <stdio.h>
```

Enter base and exponent: 2 5

```
#include <math.h>
```

2.00 ^ 5.00 = 32.00

```
int main() {
```

```
    double base, exp, result;
```

```
    printf("Enter base and exponent: ");
```

```
    scanf("%lf %lf", &base, &exp);
```

```
    result = pow(base, exp);
```

```
    printf("%.2lf ^ %.2lf = %.2lf", base, exp, result);
```

```
    return 0;
```

```
}
```

1 8 Check Character is Vowel or Consonant

```
#include <stdio.h>
int main() {
    char ch;
    printf("Enter a character: ");
    scanf(" %c", &ch); // space before %c to ignore newline

    if (ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u' ||
        ch=='A' || ch=='E' || ch=='I' || ch=='O' || ch=='U')
        printf("%c is a vowel", ch);
    else
        printf("%c is a consonant", ch);
    return 0;
}
```

Enter a character: E

E is a vowel

1 9 Check Alphabet, Digit, or Special Character

```
#include <stdio.h>
int main() {
    char ch;
    printf("Enter a character: ");
    scanf(" %c", &ch);

    if ((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z'))
        printf("Alphabet");
    else if (ch >= '0' && ch <= '9')
        printf("Digit");
    else
        printf("Special Character");

    return 0;
}
```

Enter a character: @
Special Character

2 0 Display Size of All Data Types

```
#include <stdio.h>
int main() {
    printf("Size of int: %lu bytes\n", sizeof(int));
    printf("Size of float: %lu bytes\n", sizeof(float));
    printf("Size of char: %lu bytes\n", sizeof(char));
    printf("Size of double: %lu bytes\n", sizeof(double));
    return 0;
}
```

Size of int: 4 bytes
Size of float: 4 bytes
Size of char: 1 bytes
Size of double: 8 bytes

2 **1** Print 1 to 10 using loop

```
#include <stdio.h>
int main() {
    for (int i = 1; i <= 10; i++) {
        printf("%d ", i);
    }
    return 0;
}
```

1 2 3 4 5 6 7 8 9 10

2 2 Sum of Even Numbers from 1 to N

```
#include <stdio.h>
int main() {
    int n, sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);

    for (int i = 2; i <= n; i += 2) {
        sum += i;
    }

    printf("Sum of even numbers = %d", sum);
    return 0;
}
```

2 **3** Sum of Digits of a Number

```
#include <stdio.h>
int main() {
    int n, sum = 0, rem;
    printf("Enter a number: ");
    scanf("%d", &n);

    while (n != 0) {
        rem = n % 10;
        sum += rem;
        n = n / 10;
    }

    printf("Sum of digits = %d", sum);
    return 0;
}
```

Enter a number: 1234

Sum of digits = 10

2 4 Reverse a Number

```
#include <stdio.h>
int main() {
    int n, rev = 0, rem;
    printf("Enter a number: ");
    scanf("%d", &n);

    while (n != 0) {
        rem = n % 10;
        rev = rev * 10 + rem;
        n = n / 10;
    }

    printf("Reversed number = %d", rev);
    return 0;
}
```


2 5 Check Palindrome Number

```
#include <stdio.h>
int main() {
    int n, temp, rev = 0, rem;
    printf("Enter a number: ");
    scanf("%d", &n);
    temp = n;

    while (n != 0) {
        rem = n % 10;
        rev = rev * 10 + rem;
        n = n / 10;
    }

    if (rev == temp)
        printf("Palindrome number");
    else
        printf("Not a palindrome");
    return 0;
}
```

2 6 Armstrong Number Check

```
#include <stdio.h>
#include <math.h>
int main() {
    int n, temp, rem, sum = 0, digits = 0;
    printf("Enter a number: ");
    scanf("%d", &n);
    temp = n;

    // Count digits
    while (temp != 0) {
        digits++;
        temp /= 10;
    }

    temp = n;
    while (temp != 0) {
        rem = temp % 10;
        sum += pow(rem, digits);
        temp /= 10;
    }

    if (sum == n)
        printf("Armstrong number");
    else
        printf("Not an Armstrong number");
    return 0;
}
```

27 Prime Number Check

```
#include <stdio.h>
int main() {
    int n, i, flag = 0;
    printf("Enter a number: ");
    scanf("%d", &n);

    if (n <= 1) flag = 1; // 0 and 1 are not prime
    for (i = 2; i <= n / 2; i++) {
        if (n % i == 0) {
            flag = 1;
            break;
        }
    }

    if (flag == 0)
        printf("Prime number");
    else
        printf("Not a prime number");

    return 0;
}
```

2 8 Print All Prime Numbers Between 1 and 100

```
#include <stdio.h>
int main() {
    int i, j, flag;
    printf("Prime numbers between 1 and 100 are:\n");

    for (i = 2; i <= 100; i++) {
        flag = 0;
        for (j = 2; j <= i / 2; j++) {
            if (i % j == 0) {
                flag = 1;
                break;
            }
        }
        if (flag == 0)
            printf("%d ", i);
    }
    return 0;
}
```

Prime numbers between 1 and 100 are:
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47
53 59 61 67 71 73 79 83 89 97

2 9 Fibonacci Series (Loop)

```
#include <stdio.h>
int main() {
    int n, a = 0, b = 1, next;
    printf("Enter number of terms: ");
    scanf("%d", &n);

    printf("Fibonacci Series: ");
    for (int i = 1; i <= n; i++) {
        printf("%d ", a);
        next = a + b;
        a = b;
        b = next;
    }
    return 0;
}
```

3 0 LCM of Two Numbers

```
#include <stdio.h>
int main() {
    int a, b, max;
    printf("Enter two numbers: ");
    scanf("%d%d", &a, &b);

    max = (a > b) ? a : b;

    while (1) {
        if (max % a == 0 && max % b == 0) {
            printf("LCM = %d", max);
            break;
        }
        ++max;
    }
    return 0;
}
```

Enter two numbers: 12 18
LCM = 36

3 1 GCD (Greatest Common Divisor) of Two Numbers

```
#include <stdio.h>
int main() {
    int a, b, gcd;
    printf("Enter two numbers: ");
    scanf("%d%d", &a, &b);

    for (int i = 1; i <= a && i <= b; i++) {
        if (a % i == 0 && b % i == 0)
            gcd = i;
    }

    printf("GCD = %d", gcd);
    return 0;
}
```

3 2 Count Number of Digits

```
#include <stdio.h>
int main() {
    int n, count = 0;
    printf("Enter a number: ");
    scanf("%d", &n);

    if (n == 0) count = 1;
    while (n != 0) {
        count++;
        n /= 10;
    }

    printf("Number of digits = %d", count);
    return 0;
}
```


3 3 Multiplication Table

```
#include <stdio.h>
```

```
int main() {
```

```
    int n;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &n);
```

```
    for (int i = 1; i <= 10; i++) {
```

```
        printf("%d x %d = %d\n", n, i, n * i);
```

```
    }
```

```
    return 0;
```

```
}
```

Enter a number: 5

5 x 1 = 5

5 x 2 = 10

...

5 x 10 = 50

3 4 Strong Number Check

A number is strong if the sum of factorials of its digits = number itself.

Example: $145 \rightarrow 1! + 4! + 5! = 145$

```
#include <stdio.h>
int main() {
    int n, temp, rem, fact, i, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &n);
    temp = n;

    while (n != 0) {
        rem = n % 10;
        fact = 1;
        for (i = 1; i <= rem; i++)
            fact *= i;
        sum += fact;
        n /= 10;
    }

    if (sum == temp)
        printf("Strong number");
    else
        printf("Not a strong number");

    return 0;
}
```

Enter a number: 145
Strong number

3 **5** Perfect Number Check

A number is perfect if sum of its proper divisors = number.

Example: $6 \rightarrow 1 + 2 + 3 = 6$

```
#include <stdio.h>
int main() {
    int n, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &n);

    for (int i = 1; i < n; i++) {
        if (n % i == 0)
            sum += i;
    }

    if (sum == n)
        printf("Perfect number");
    else
        printf("Not a perfect number");

    return 0;
}
```

3 6 Sum of Series (1 + 2 + 3 + ... + n)

```
#include <stdio.h>
int main() {
    int n, sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);

    for (int i = 1; i <= n; i++)
        sum += i;

    printf("Sum of series = %d", sum);
    return 0;
}
```

3 7 Pattern – Half Pyramid of Stars

```
*  
* *  
* * *  
* * * *
```

```
#include <stdio.h>  
int main() {  
    int n;  
    printf("Enter number of rows: ");  
    scanf("%d", &n);  
  
    for (int i = 1; i <= n; i++) {  
        for (int j = 1; j <= i; j++) {  
            printf("* ");  
        }  
        printf("\n");  
    }  
    return 0;  
}
```

3 8 Pattern – Inverted Pyramid

```
* * * *  
* * *  
* *  
*
```

```
#include <stdio.h>  
int main() {  
    int n;  
    printf("Enter number of rows: ");  
    scanf("%d", &n);  
  
    for (int i = n; i >= 1; i--) {  
        for (int j = 1; j <= i; j++) {  
            printf("* ");  
        }  
        printf("\n");  
    }  
    return 0;  
}
```

3 9 Pattern – Number Pyramid

```
#include <stdio.h>
int main() {
    int n;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            printf("%d ", j);
        }
        printf("\n");
    }
    return 0;
}
```

4 0 Pattern – Floyd's Triangle

```
#include <stdio.h>
int main() {
    int n, num = 1;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            printf("%d ", num);
            num++;
        }
        printf("\n");
    }
    return 0;
}
```


41. Find Factorial using Function

```
#include <stdio.h>
```

```
int factorial(int n) {  
    int i, fact = 1;  
    for(i = 1; i <= n; i++)  
        fact *= i;  
    return fact;  
}
```

```
int main() {  
    int num;  
    printf("Enter a number: ");  
    scanf("%d", &num);  
    printf("Factorial = %d", factorial(num));  
    return 0;  
}
```

42. Check Prime using Function

```
#include <stdio.h>

int isPrime(int n) {
    if (n <= 1) return 0;
    for (int i = 2; i <= n / 2; i++)
        if (n % i == 0)
            return 0;
    return 1;
}

int main() {
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);
    if (isPrime(num))
        printf("Prime Number");
    else
        printf("Not Prime");
    return 0;
}
```

43. Power of Number using Recursion

```
#include <stdio.h>
```

```
int power(int base, int exp) {  
    if (exp == 0)  
        return 1;  
    else  
        return base * power(base, exp - 1);  
}
```

```
int main() {  
    int base, exp;  
    printf("Enter base and exponent: ");  
    scanf("%d %d", &base, &exp);  
    printf("Result = %d", power(base, exp));  
    return 0;  
}
```

Enter base and exponent: 2 5
Result = 32

44. Fibonacci using Recursion

```
#include <stdio.h>

int fibonacci(int n) {
    if (n == 0) return 0;
    if (n == 1) return 1;
    return fibonacci(n - 1) + fibonacci(n - 2);
}

int main() {
    int n, i;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    printf("Fibonacci Series: ");
    for (i = 0; i < n; i++)
        printf("%d ", fibonacci(i));
    return 0;
}
```

45. Sum of Array Elements

```
#include <stdio.h>

int main() {
    int n, i, sum = 0;
    printf("Enter size of array: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter elements: ");
    for(i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
        sum += arr[i];
    }
    printf("Sum = %d", sum);
    return 0;
}
```

46. Find Largest Element in Array

```
#include <stdio.h>

int main() {
    int n, i, largest;
    printf("Enter size of array: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter elements: ");
    for(i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    largest = arr[0];
    for(i = 1; i < n; i++)
        if(arr[i] > largest)
            largest = arr[i];
    printf("Largest element = %d", largest);
    return 0;
}
```

47. Find Smallest Element in Array

```
#include <stdio.h>

int main() {
    int n, i, smallest;
    printf("Enter size of array: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter elements: ");
    for(i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    smallest = arr[0];
    for(i = 1; i < n; i++)
        if(arr[i] < smallest)
            smallest = arr[i];
    printf("Smallest element = %d", smallest);
    return 0;
}
```

48. Sort Array in Ascending Order

```
#include <stdio.h>

int main() {
    int n, i, j, temp;
    printf("Enter size of array: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter elements: ");
    for(i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    for(i = 0; i < n-1; i++)
        for(j = i+1; j < n; j++)
            if(arr[i] > arr[j]) {
                temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
    printf("Sorted array: ");
    for(i = 0; i < n; i++)
        printf("%d ", arr[i]);
    return 0;
}
```


49. Reverse an Array

```
#include <stdio.h>

int main() {
    int n, i;
    printf("Enter size of array: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter elements: ");
    for(i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    printf("Reversed array: ");
    for(i = n - 1; i >= 0; i--)
        printf("%d ", arr[i]);
    return 0;
}
```

50. Count Even & Odd Numbers in Array

```
#include <stdio.h>

int main() {
    int n, i, even = 0, odd = 0;
    printf("Enter size of array: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter elements: ");
    for(i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
        if(arr[i] % 2 == 0)
            even++;
        else
            odd++;
    }
    printf("Even = %d, Odd = %d", even, odd);
    return 0;
}
```

61. Find length of string

```
#include <stdio.h>

int main() {
    char str[100];
    int i, length = 0;

    printf("Enter a string: ");
    gets(str); // input string

    for(i = 0; str[i] != '\0'; i++) {
        length++;
    }

    printf("Length of string = %d", length);
    return 0;
}
```

62. Reverse a string

```
#include <stdio.h>
#include <string.h>

int main() {
    char str[100], rev[100];
    int i, len, j = 0;

    printf("Enter a string: ");
    gets(str);

    len = strlen(str);

    for(i = len - 1; i >= 0; i--) {
        rev[j++] = str[i];
    }
    rev[j] = '\0';

    printf("Reversed string: %s", rev);
    return 0;
}
```

63. Check palindrome string

```
#include <stdio.h>
#include <string.h>

int main() {
    char str[100];
    int i, len, flag = 0;

    printf("Enter a string: ");
    gets(str);

    len = strlen(str);

    for(i = 0; i < len / 2; i++) {
        if(str[i] != str[len - i - 1]) {
            flag = 1;
            break;
        }
    }

    if(flag == 0)
        printf("String is Palindrome");
    else
        printf("String is NOT Palindrome");

    return 0;
}
```

64. Concatenate two strings (without strcat)

```
#include <stdio.h>

int main() {
    char str1[100], str2[100];
    int i, j;

    printf("Enter first string: ");
    gets(str1);
    printf("Enter second string: ");
    gets(str2);

    for(i = 0; str1[i] != '\0'; i++); // move to end of str1

    for(j = 0; str2[j] != '\0'; j++, i++) {
        str1[i] = str2[j];
    }
    str1[i] = '\0';

    printf("Concatenated String: %s", str1);
    return 0;
}
```

65. Copy string (without strcpy)

```
#include <stdio.h>

int main() {
    char str1[100], str2[100];
    int i;

    printf("Enter a string: ");
    gets(str1);

    for(i = 0; str1[i] != '\0'; i++) {
        str2[i] = str1[i];
    }
    str2[i] = '\0';

    printf("Copied String: %s", str2);
    return 0;
}
```

```
#include <stdio.h>
#include <ctype.h>
```

```
int main() {
    char str[100];
    int vowels = 0, consonants = 0, i;
    printf("Enter a string: ");
    gets(str);

    for(i = 0; str[i] != '\0'; i++) {
        char ch = tolower(str[i]);
        if(ch >= 'a' && ch <= 'z') {
            if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u')
                vowels++;
            else
                consonants++;
        }
    }
    printf("Vowels = %d, Consonants = %d", vowels,
consonants);
    return 0;
}
```

66. Count Vowels and Consonants

67. Count Digits, Alphabets, and Spaces

```
#include <stdio.h>
#include <ctype.h>

int main() {
    char str[100];
    int digits=0, alphabets=0, spaces=0, i;
    printf("Enter a string: ");
    gets(str);

    for(i=0; str[i]!='\0'; i++) {
        if(isdigit(str[i]))
            digits++;
        else if(isalpha(str[i]))
            alphabets++;
        else if(str[i]==' ')
            spaces++;
    }
    printf("Alphabets=%d, Digits=%d, Spaces=%d", alphabets, digits, spaces);
    return 0;
}
```

68. Convert String to Uppercase

```
#include <stdio.h>
#include <ctype.h>

int main() {
    char str[100];
    int i;
    printf("Enter a string: ");
    gets(str);

    for(i=0; str[i]!='\0'; i++)
        str[i] = toupper(str[i]);

    printf("Uppercase: %s", str);
    return 0;
}
```

69. Convert String to Lowercase

```
#include <stdio.h>
#include <ctype.h>

int main() {
    char str[100];
    int i;
    printf("Enter a string: ");
    gets(str);

    for(i=0; str[i]!='\0'; i++)
        str[i] = tolower(str[i]);

    printf("Lowercase: %s", str);
    return 0;
}
```

70. Compare Two Strings

```
#include <stdio.h>
#include <string.h>

int main() {
    char str1[100], str2[100];
    printf("Enter first string: ");
    gets(str1);
    printf("Enter second string: ");
    gets(str2);

    if(strcmp(str1, str2) == 0)
        printf("Strings are equal");
    else
        printf("Strings are not equal");
    return 0;
}
```

```
#include <stdio.h>
#include <string.h>
```

```
int main() {
    char str[100], temp;
    int i, j;
    printf("Enter a string: ");
    gets(str);
    int len = strlen(str);

    for(i=0; i<len-1; i++)
        for(j=i+1; j<len; j++)
            if(str[i] > str[j]) {
                temp = str[i];
                str[i] = str[j];
                str[j] = temp;
            }

    printf("Sorted string: %s", str);
    return 0;
}
```

71. Sort Characters of String

72. Frequency of Characters in String

```
#include <stdio.h>
#include <string.h>

int main() {
    char str[100];
    int freq[256] = {0}, i;
    printf("Enter a string: ");
    gets(str);

    for(i=0; str[i]!='\0'; i++)
        freq[(unsigned char)str[i]]++;

    printf("Character frequencies:\n");
    for(i=0; i<256; i++)
        if(freq[i] != 0)
            printf("%c = %d\n", i, freq[i]);
    return 0;
}
```

73. Remove Spaces from String

```
#include <stdio.h>

int main() {
    char str[100];
    int i, j=0;
    printf("Enter a string: ");
    gets(str);

    for(i=0; str[i]!='\0'; i++)
        if(str[i] != ' ')
            str[j++] = str[i];
    str[j] = '\0';

    printf("String without spaces: %s", str);
    return 0;
}
```

74. Find Substring in String

```
#include <stdio.h>
#include <string.h>

int main() {
    char str[100], sub[50];
    printf("Enter main string: ");
    gets(str);
    printf("Enter substring: ");
    gets(sub);

    if(strstr(str, sub))
        printf("Substring found!");
    else
        printf("Substring not found!");
    return 0;
}
```


75. Replace Character in String

```
#include <stdio.h>

int main() {
    char str[100], oldChar, newChar;
    printf("Enter string: ");
    gets(str);
    printf("Enter character to replace: ");
    scanf("%c", &oldChar);
    getchar();
    printf("Enter new character: ");
    scanf("%c", &newChar);

    for(int i=0; str[i]!='\0'; i++)
        if(str[i]==oldChar)
            str[i]=newChar;

    printf("Modified string: %s", str);
    return 0;
}
```

76. Count Words in Sentence

```
#include <stdio.h>

int main() {
    char str[100];
    int count = 1, i;
    printf("Enter a sentence: ");
    gets(str);

    for(i=0; str[i]!='\0'; i++)
        if(str[i]==' ' && str[i+1]!=' ' && str[i+1]!='\0')
            count++;

    printf("Total words = %d", count);
    return 0;
}
```

```

#include <stdio.h>
#include <string.h>

void reverse(char *start, char *end) {
    char temp;
    while(start < end) {
        temp = *start;
        *start++ = *end;
        *end-- = temp;
    }
}

int main() {
    char str[200], *word_start = NULL;
    printf("Enter a sentence: ");
    gets(str);

    for(int i=0; ; i++) {
        if(word_start == NULL && str[i] != ' ' && str[i] != '\0')
            word_start = &str[i];
        if((str[i] == ' ' || str[i] == '\0') && word_start) {
            reverse(word_start, &str[i-1]);
            word_start = NULL;
        }
        if(str[i] == '\0') break;
    }
    printf("Reversed each word: %s", str);
    return 0;
}

```

77. Reverse Each Word in Sentence

78. Print String Without Vowels

```
#include <stdio.h>
#include <ctype.h>

int main() {
    char str[100];
    printf("Enter string: ");
    gets(str);

    printf("Without vowels: ");
    for(int i=0; str[i]!='\0'; i++) {
        char ch = tolower(str[i]);
        if(ch!='a' && ch!='e' && ch!='i' && ch!='o' && ch!='u')
            printf("%c", str[i]);
    }
    return 0;
}
```

79. Check Anagram

```
#include <stdio.h>
#include <string.h>
```

```
int main() {
    char str1[100], str2[100];
    int freq[256] = {0}, i;

    printf("Enter first string: ");
    gets(str1);
    printf("Enter second string: ");
    gets(str2);

    if(strlen(str1) != strlen(str2)) {
        printf("Not an Anagram");
        return 0;
    }

    for(i=0; str1[i]!='\0'; i++) {
        freq[(unsigned char)str1[i]]++;
        freq[(unsigned char)str2[i]]--;
    }

    for(i=0; i<256; i++)
        if(freq[i]!=0) {
            printf("Not an Anagram");
            return 0;
        }

    printf("Strings are Anagrams");
    return 0;
}
```

80. Convert String to Integer

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    char str[50];
    printf("Enter numeric string: ");
    gets(str);

    int num = atoi(str);
    printf("Integer value = %d", num);
    return 0;
}
```

81. Pointer Basic Example

```
#include <stdio.h>
int main() {
    int a = 10;
    int *p = &a;
    printf("Value of a = %d\n", a);
    printf("Address of a = %p\n", p);
    printf("Value using pointer = %d", *p);
    return 0;
}
```

82. Add Two Numbers using Pointers

```
#include <stdio.h>
int main() {
    int a, b, sum;
    int *p1 = &a, *p2 = &b;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    sum = *p1 + *p2;
    printf("Sum = %d", sum);
    return 0;
}
```


83. Swap using Pointers

```
#include <stdio.h>
void swap(int *x, int *y) {
    int temp = *x;
    *x = *y;
    *y = temp;
}
int main() {
    int a, b;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    swap(&a, &b);
    printf("After swap: a=%d, b=%d", a, b);
    return 0;
}
```

84. Access Array using Pointers

```
#include <stdio.h>
int main() {
    int arr[5] = {10, 20, 30, 40, 50};
    int *p = arr;
    for(int i = 0; i < 5; i++)
        printf("%d ", *(p + i));
    return 0;
}
```

85. Sum of Array using Pointer

```
#include <stdio.h>
int main() {
    int n, sum = 0;
    printf("Enter size: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter elements: ");
    for(int i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    int *p = arr;
    for(int i = 0; i < n; i++)
        sum += *(p + i);
    printf("Sum = %d", sum);
    return 0;
}
```

86. Pointer to Pointer Example

```
#include <stdio.h>
int main() {
    int a = 10;
    int *p = &a;
    int **q = &p;
    printf("Value of a = %d\n", a);
    printf("Value using *p = %d\n", *p);
    printf("Value using **q = %d", **q);
    return 0;
}
```

87. Function using Pointer (Call by Reference)

```
#include <stdio.h>
void add(int *x, int *y, int *s) {
    *s = *x + *y;
}
int main() {
    int a, b, sum;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    add(&a, &b, &sum);
    printf("Sum = %d", sum);
    return 0;
}
```

88. Structure – Student Record (Name, Roll, Marks)

```
#include <stdio.h>
struct Student {
    char name[50];
    int roll;
    float marks;
};
int main() {
    struct Student s;
    printf("Enter name, roll, marks: ");
    scanf("%s %d %f", s.name, &s.roll, &s.marks);
    printf("\nName: %s\nRoll: %d\nMarks: %.2f", s.name, s.roll, s.marks);
    return 0;
}
```

89. Structure Array Example

```
#include <stdio.h>
struct Student {
    char name[50];
    int roll;
    float marks;
};
int main() {
    struct Student s[3];
    for(int i=0; i<3; i++) {
        printf("Enter name, roll, marks for student %d: ", i+1);
        scanf("%s %d %f", s[i].name, &s[i].roll, &s[i].marks);
    }
    printf("\nStudent Records:\n");
    for(int i=0; i<3; i++)
        printf("%s %d %.2f\n", s[i].name, s[i].roll, s[i].marks);
    return 0;
}
```

90. Pass Structure to Function

```
#include <stdio.h>
struct Student {
    char name[50];
    int roll;
    float marks;
};
void display(struct Student s) {
    printf("\nName: %s\nRoll: %d\nMarks: %.2f", s.name, s.roll, s.marks);
}
int main() {
    struct Student s = {"Amit", 1, 89.5};
    display(s);
    return 0;
}
```


91. Nested Structure Example

```
#include <stdio.h>
struct Address {
    char city[30];
    int pincode;
};
struct Student {
    char name[30];
    int roll;
    struct Address addr;
};
int main() {
    struct Student s = {"Ravi", 10, {"Pune", 411001}};
    printf("Name: %s\nRoll: %d\nCity: %s\nPincode: %d",
        s.name, s.roll, s.addr.city, s.addr.pincode);
    return 0;
}
```

92. Union Example

```
#include <stdio.h>
union Data {
    int i;
    float f;
    char str[20];
};
int main() {
    union Data d;
    d.i = 10;
    printf("i = %d\n", d.i);
    d.f = 20.5;
    printf("f = %.2f\n", d.f);
    printf("i after float = %d", d.i);
    return 0;
}
```

93. Enum Example

```
#include <stdio.h>
enum Week {Mon, Tue, Wed, Thu, Fri, Sat, Sun};
int main() {
    enum Week today;
    today = Wed;
    printf("Day number = %d", today);
    return 0;
}
```

94. File – Write to File

```
#include <stdio.h>
int main() {
    FILE *fp;
    fp = fopen("data.txt", "w");
    if(fp == NULL) {
        printf("Error opening file!");
        return 1;
    }
    fprintf(fp, "Hello File Handling in C");
    fclose(fp);
    printf("Data written successfully!");
    return 0;
}
```

95. File – Read from File

```
#include <stdio.h>
int main() {
    FILE *fp;
    char ch;
    fp = fopen("data.txt", "r");
    if(fp == NULL) {
        printf("Error opening file!");
        return 1;
    }
    while((ch = fgetc(fp)) != EOF)
        putchar(ch);
    fclose(fp);
    return 0;
}
```

96. Count Characters in File

```
#include <stdio.h>
int main() {
    FILE *fp;
    char ch;
    int count = 0;
    fp = fopen("data.txt", "r");
    while((ch = fgetc(fp)) != EOF)
        count++;
    fclose(fp);
    printf("Total characters = %d", count);
    return 0;
}
```

97. Copy Content of One File to Another

```
#include <stdio.h>
int main() {
    FILE *f1, *f2;
    char ch;
    f1 = fopen("source.txt", "r");
    f2 = fopen("dest.txt", "w");
    while((ch = fgetc(f1)) != EOF)
        fputc(ch, f2);
    fclose(f1);
    fclose(f2);
    printf("File copied successfully!");
    return 0;
}
```

98. Append Data to File

```
#include <stdio.h>
int main() {
    FILE *fp;
    fp = fopen("data.txt", "a");
    fprintf(fp, "\nThis is appended text.");
    fclose(fp);
    printf("Data appended successfully!");
    return 0;
}
```


99. Count Lines, Words, and Characters in File

```
#include <stdio.h>
#include <ctype.h>
int main() {
    FILE *fp;
    char ch;
    int lines = 0, words = 0, chars = 0, inWord = 0;
    fp = fopen("data.txt", "r");
    while((ch = fgetc(fp)) != EOF) {
        chars++;
        if(ch == '\n')
            lines++;
        if(isspace(ch))
            inWord = 0;
        else if(!inWord) {
            inWord = 1;
            words++;
        }
    }
    fclose(fp);
    printf("Lines=%d Words=%d Characters=%d", lines, words, chars);
    return 0;
}
```

100. Student Record Management (Mini Project)

```
#include <stdio.h>
#include <stdlib.h>
```

```
struct Student {
    int roll;
    char name[50];
    float marks;
};
```

```
int main() {
    struct Student s;
    FILE *fp;
    int choice;
```

```
while(1) {
    printf("\n1.Add Record\n2.Display Records\n3.Exit\nEnter
choice: ");
    scanf("%d", &choice);

    if(choice == 1) {
        fp = fopen("student.dat", "a");
        printf("Enter Roll, Name, Marks: ");
        scanf("%d %s %f", &s.roll, s.name, &s.marks);
        fwrite(&s, sizeof(s), 1, fp);
        fclose(fp);
    }
    else if(choice == 2) {
        fp = fopen("student.dat", "r");
        printf("\nRoll\tName\tMarks\n");
        while(fread(&s, sizeof(s), 1, fp))
            printf("%d\t%s\t%.2f\n", s.roll, s.name, s.marks);
        fclose(fp);
    }
    else if(choice == 3)
        exit(0);
    else
        printf("Invalid choice!");
}
}
```